

they hit on a Harry Potter string of super-best-sellers – provided of course that they publish quality work that has a small probability of being of very high appeal.

In the markets, there is a category of traders who have *inverse* rare events, for whom volatility is often a bearer of good news. These traders lose money frequently, but in small amounts, and make money rarely, but in large amounts. I call them crisis hunters. I am happy to be one of them.

WHY DON'T STATISTICIANS DETECT RARE EVENTS?

Statistics to the layman can appear rather complex, but the concept behind what is used today is so simple that my French mathematician friends call it deprecatorily “cuisine”. It is all based on one simple notion; the more information you have, the more you are confident about the outcome. Now the problem; by how much? Common statistical method is based on the steady augmentation of the confidence level, in nonlinear proportion to the number of observations. That is, for an n times increase in the sample size, we increase our knowledge by the square root of n . Suppose I am drawing from an urn containing red and black balls. My confidence level about the relative proportion of red and black balls, after 20 drawings is not twice the one I have after 10 drawings; it is merely multiplied by the square root of 2 (that is, 1.41).

Where statistics becomes complicated, and fails us, is when we have distributions that are not symmetric, like the urn above. If there is a very small probability of finding a red ball in an urn dominated by black ones, then our knowledge about the *absence* of red balls will increase very slowly – more slowly than at the expected square root of n rate. On the other hand our knowledge of the *presence* of red balls will dramatically improve once one of them is found. This asymmetry in knowledge is not trivial; it is central in this book – it is a central philosophical problem for such people as Hume and Karl Popper (on that, later).

To assess an investor's performance, we either need more astute, and less intuitive, techniques, or we may have to limit our assessments to situations where our judgment is independent of the frequency of these events.

A MISCHIEVOUS CHILD REPLACES THE BLACK BALLS

But there is even worse news. In some cases, if the incidence of red balls is itself randomly distributed, we will never get to know the composition of the urn. This is called *the problem of stationarity*. Think of an urn that is hollow at the bottom. As I am sampling from it, and without my being aware of it, some mischievous child is adding balls of one color or another. My inference thus becomes insignificant. I may infer that the red balls represent 50% of the urn while the mischievous child, hearing me, would swiftly replace all the red balls with black ones. This makes much of our knowledge derived through statistics quite shaky.

The very same effect takes place in the market. We take past history as a single homogeneous sample and believe that we have considerably increased our knowledge of the future from the observation of the sample of the past. What if vicious children were changing the composition of the urn? In other words, what if things have changed?

I have studied and practiced econometrics for more than half my life (since I was 19), both in the classroom and in the activity of a quantitative derivatives trader. The “science” of econometrics consists of the application of statistics to samples taken at different periods of time, which we called *time series*. It is based on studying the time series of economic variables, data, and other matters. In the beginning, when I knew close to nothing (that is even less than today), I wondered whether the time series reflecting the activity of people now dead or retired should matter for predicting the future. Econometricians who knew a lot more than I did about these matters asked no such question; this hinted that it was in all likelihood a stupid inquiry. One prominent econometrician, Hashem Pesaran, answered a similar question by recommending to do “more and better econometrics”. I am now convinced that, perhaps, most of econometrics could be useless – much of what financial statisticians know would not be worth knowing. For a sum of zeros, even repeated a billion times, remains zero; likewise an accumulation of research and gains in complexity will lead to naught if there is no firm ground beneath it. Studying the European markets of the 1990s will certainly be of great help to a historian; but what kind of inference can we make now that the structure of the institutions and the markets has changed so much?